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LEVERAGED POSITIONS

Learning Outcome

- calculate and interpret the leverage ratio, the rate of return on a margin transaction, and the security price at which the investor would receive a margin call

In many markets, traders can buy securities by borrowing some of the purchase price. They usually borrow the money from their brokers. The borrowed money is called the **margin loan**, and they are said to buy on margin. The interest rate that the buyers pay for their margin loan is called the **call money rate**. The call money rate is above the government bill rate and is negotiable. Large buyers generally obtain more favorable rates than do retail buyers. For institutional-size buyers, the call money rate is quite low because the loans are generally well secured by securities held as collateral by the lender.

Trader's equity is that portion of the security price that the buyer must supply. Traders who buy securities on margin are subject to minimum margin requirements. The **initial margin requirement** is the minimum fraction of the purchase price that must be trader's equity. This requirement may be set by the government, the exchange, or the exchange clearinghouse. For example, in the United States, the Federal Reserve Board sets the initial margin requirement through Regulation T. In Hong Kong SAR, the Securities and Futures Commission sets the margin requirements. In all markets, brokers often require more equity than the government-required minimum from their clients when lending to them.

Many markets allow brokers to lend their clients more money if the brokers use risk models to measure and control the overall risk of their clients' portfolios. This system is called portfolio margining.

Buying securities on margin can greatly increase the potential gains or losses for a given amount of equity in a position because the trader can buy more securities on margin than he could otherwise. The buyer thus earns greater profits when prices rise and suffers greater losses when prices fall. The relation between risk and borrowing is called **financial leverage** (often simply called leverage). Traders leverage their positions when they borrow to buy more securities. A highly leveraged position is large relative to the equity that supports it.

The leverage ratio is the ratio of the value of the position to the value of the equity investment in it. The leverage ratio indicates how many times larger a position is than the equity that supports it. The maximum leverage ratio associated with a position financed by the minimum margin requirement is one divided by the minimum margin requirement. If the requirement is 40 percent, then the maximum leverage ratio is 2.5 = 100% position ÷ 40% equity.

The leverage ratio indicates how much more risky a leveraged position is relative to an unleveraged position. For example, if a stock bought on 40 percent margin rises 10 percent, the buyer will experience a 25 percent (2.5 × 10%) return on the equity investment in her leveraged position. But if the stock falls by 10 percent, the return on the equity investment will be −25 percent (before the interest on the margin loan and before payment of commissions).

Financial analysts must be able to compute the total return to the equity investment in a leveraged position. The total return depends on the price change of the purchased security, the dividends or interest paid by the security, the interest paid on the margin loan,

EXAMPLE 19

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Computing Total Return to a Leveraged Stock Purchase

A buyer buys stock on margin and holds the position for exactly one year, during which time the stock pays a dividend. For simplicity, assume that the interest on the loan and the dividend are both paid at the end of the year.

Purchase price	$20/share
Sale price	$15/share
Shares purchased	1,000
Leverage ratio	2.5
Call money rate	5%
Dividend	$0.10/share
Commission	$0.01/share

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1. What is the total return on this investment?

Hide Solution

Solution to 1:

To find the return on this investment, first determine the initial equity and then determine the equity remaining after the sale. The total purchase price is $20,000. The leverage ratio of 2.5 indicates that the buyer's equity financed 40 percent = (1 ÷ 2.5) of the purchase price. Thus, the equity investment is $8,000 = 40% of $20,000. The $12,000 remainder is borrowed. The actual investment is slightly higher because the buyer must pay a commission of $10 = $0.01/share × 1,000 shares to buy the stock. The total initial investment is $8,010.

At the end of the year, the stock price has declined by $5/share. The buyer lost $5,000 = $5/share × 1,000 shares as a result of the price change. In addition, the buyer has to pay interest at 5 percent on the $12,000 loan, or $600. The buyer also receives a dividend of $0.10/share, or $100. The trader's equity remaining after the sale is computed from the initial equity investment as follows:

Initial investment	$8,010
Purchase commission	–10
Trading gains/losses	–5,000
Margin interest paid	–600
Dividends received	100
Sales commission paid	–10
Remaining equity	$2,490

or

Proceeds on sale	$15,000
Payoff loan	–12,000
Margin interest paid	–600
Dividends received	100
Sales commission paid	–10
Remaining equity	$2,490

Hide Solution

Solution to 2:

The realized loss is substantially greater than the stock price return of ($15 – $20)/$20 = –25%. Most of the difference is because of the leverage with the remainder primarily the result of the interest paid on the loan. Based on the leverage alone and ignoring the other cash flows, we would expect that the return on the equity would be –62.5% = 2.5 leverage times the –25% stock price return.

In the above example, if the stock dropped more than the buyer's original 40 percent margin (ignoring commissions, interest, and dividends), the trader's equity would have become negative. In that case, the investor would owe his broker more than the stock is worth. Brokers often lose money in such situations if the buyer does not repay the loan out of other funds.

To prevent such losses, brokers require that margin buyers always have a minimum amount of equity in their positions. This minimum is called the **maintenance margin requirement**. It is usually 25 percent of the current value of the position, but it may be higher or lower depending on the volatility of the instrument and the policies of the broker.

If the value of the equity falls below the maintenance margin requirement, the buyer will receive a **margin call**, or request for additional equity. If the buyer does not deposit additional equity with the broker in a timely manner, the broker will close the position to prevent further losses and thereby secure repayment of the margin loan.

When you buy securities on margin, you must know the price at which you will receive a margin call if prices drop. The answer to this question depends on your initial equity and on the maintenance margin requirement.

EXAMPLE 20

Margin Call Price

1. A trader buys stock on margin posting 40 percent of the initial stock price of $20 as equity. The maintenance margin requirement for the position is 25 percent. Below what price will a margin call occur?

Hide Solution

Solution:

The trader's initial equity is 40 percent of the initial stock price of $20, or $8 per share. Subsequent changes in equity per share are equal to the share price change so that equity per share is equal to $8 + ($P$ – 20) where P is the current share price. The margin call takes place when equity drops below the 25 percent maintenance margin requirement. The price below which a margin call will take place is the solution to the following equation:

$$\frac{\text{Equity/share}}{\text{Price/share}} = \frac{\$8 + P - 20}{P} = 25\%$$

which occurs at P = 16. When the price drops below $16, the equity will be under $4/share, which is less than 25 percent of the price.

Traders who sell securities short are also subject to margin requirements because they have borrowed securities. Initially, the trader's equity supporting the short position must be

buy the security back to cover the short position to prevent further losses and thereby secure repayment of the stock loan.

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Discussion

What do you want to discuss... (required)

SK

The explanation of the "maintenance margin requirement" is crap and so is the Example 20.
It says that the "maintenance margin requirement" is usually 25% of the "current value of the position". -> What exactly is the "current value of the position"? And what is it based on the example 20?
Is it 25% of $20 or 25% of the initial equity of $8?
Also in the calculation provided in the example 20: ($8 + P - 20) / P = 0,25 where P = 16, but where does $4/share values come from?
According to the text "maintenance margin requirement" based on the "current value" - is it 25% of $20, which should trigger a margin call when investors equity hits $5/share?
What did I miss in this lesson that made me loose so much time over this and being wrong?

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SK

I believe there are 2 mistakes in the calculation of the Example 19:
A) There is a mistake in the first calculation, initially adding and than deducting the commission for purchase of shares ($10).
B) There is also a mistake in the alternative way to calculate: The "commission for purchase" is not mentioned, therefore I think the remaining equity actually is $2.480

I don't understand why a commission of $10 paid for purchase of shares should increase the "equity" in any way? I don't invest the $10 into shares, I pay it to the exchange for providing me the service, or am I wrong?

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AK

In the example 19, why is the initial investement 8010 and then again the purchase commission is deducted, as 8010 includes the 10 of initial purchase comission paid?

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